



NEW GENERATION UNIVERSITY COLLEGE

Introduction to Artificial Intelligence

Individual Assignment

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Question 1: What is the difference between AI, Machine Learning, and Deep Learning? Explain with real-life examples.

Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) are often used interchangeably, but they represent different levels of a single, nested concept. AI is the broadest term and refers to any technique that enables a computer to mimic human intelligence, including reasoning, problem-solving, understanding language, or perception. Early AI systems were often built using hand-coded rules. For example, a rule-based chess program or a simple chatbot that responds with pre-written answers to specific keywords is AI, even though it does not "learn" anything.

Machine Learning is a subset of AI in which the system learns patterns directly from data instead of following rules written by a programmer. The performance of an ML system improves as it is exposed to more data. A real-life example is the spam filter in an email service: instead of a human writing rules for every possible spam message, the system is trained on thousands of examples of spam and non-spam emails and learns to recognise the patterns that distinguish them. Another example is a bank using ML to predict whether a loan applicant is likely to default, based on past repayment records.

Deep Learning is a specialised subset of Machine Learning that uses artificial neural networks with many layers (hence "deep") to automatically learn complex patterns and representations from large amounts of data. Deep Learning is especially powerful for unstructured data such as images, audio, and text. A real-life example is the facial recognition feature used to unlock smartphones, or voice assistants such as Google Assistant and Siri, which use deep neural networks to convert speech into text and understand intent. Self-driving cars also rely on deep learning to recognise pedestrians, road signs, and other vehicles from camera images.

In summary, every Deep Learning system is a Machine Learning system, and every Machine Learning system is an AI system, but not every AI system uses Machine Learning, and not every Machine Learning system uses Deep Learning. The relationship can be thought of as concentric circles: AI is the outer circle, ML is inside it, and DL is the innermost circle.

Question 2: Why is data important in Artificial Intelligence systems? What problems can arise if data is poor or biased?

Data is the foundation on which modern AI systems are built. Unlike traditional software, where a programmer explicitly defines the logic, machine learning and deep learning models learn their behaviour by finding patterns in data. The quality, quantity, and diversity of the data used for training directly determine how accurate, fair, and useful the resulting system will be. Without sufficient and representative data, even the most sophisticated algorithm cannot produce reliable results, because the model has nothing meaningful to learn from. Problems caused by poor or biased data:

Inaccurate predictions: If the training data contains errors, noise, or is incomplete, the model will learn incorrect patterns and make poor predictions in real use. For example, a crop disease detection model trained on blurry or mislabeled images may misclassify healthy plants as diseased. **Bias and discrimination:** If certain groups are underrepresented in the data, the model may perform poorly for those groups. A well-known example is facial recognition systems that were trained mostly on lighter-skinned faces and performed significantly worse for people with darker skin tones.

Poor generalisation: A model trained on data from one context (for example, a different country's traffic patterns or farming conditions) may fail when applied to a new context, such as Ethiopian roads or crops, because the underlying patterns differ.

Reinforcing existing inequalities: If historical data reflects past discrimination (for example, in hiring or lending decisions), an AI model trained on that data may learn and even amplify those same biases, leading to unfair outcomes for certain groups.

Loss of trust and harmful decisions: When an AI system makes consistently wrong or unfair decisions because of poor data, people lose trust in the technology, and in sensitive areas such as healthcare or finance, biased data can directly harm people's lives and livelihoods.

For these reasons, careful data collection, cleaning, labelling, and checking for representativeness are essential steps before any AI model is trained and deployed.

Question 3: Compare Breadth First Search (BFS) and Depth First Search (DFS) in terms of completeness, memory usage, and optimality

BFS and DFS are two of the basic uninformed search strategies in AI, and the real difference between them comes down to the order in which nodes get expanded. BFS moves through the tree level by level, while DFS picks a branch and follows it all the way down before backing up to try another. That one difference ends up affecting completeness, memory use, and whether the search actually finds the best solution.

In terms of completeness, BFS will always find a solution if one exists, since it never skips a level, it just might take a while in a big tree. DFS is only complete if the tree is finite, though. If it heads down a branch that goes on forever, it can get stuck there and never reach the part of the tree where the actual solution is.

Memory usage is where the two really differ. BFS has to keep every node on the current level in memory, and that number can grow exponentially the deeper you go ($O(b^d)$). DFS is much lighter on memory since it only has to remember the path it's currently on, so memory use grows linearly with depth ($O(b \cdot d)$).

As for optimality, BFS is optimal as long as every step costs the same, because it reaches shallower nodes before deeper ones, so the first solution it finds is the shortest. DFS doesn't have that guarantee, it can land on a solution that's much deeper (and costlier) than necessary, simply because it committed to that branch early and didn't check the others first.

Question 4: Explain heuristic search. Why is it important in AI problem solving? Give an example of where it could be useful in Ethiopian logistics or transportation.

A heuristic search is a search strategy that uses additional knowledge about the problem, known as a heuristic function, to estimate how close a given state is to the goal, and to guide the search toward more promising paths first. Unlike blind (uninformed) searches such as BFS and DFS, which explore the search space without any sense of direction, heuristic searches such as Greedy Best-First Search and A* use this estimate to prioritise which node to expand next.

Heuristic search is important in AI problem solving because real-world search spaces are often enormous, and exploring every possibility (as blind search does) would take an impractical amount of time and memory. By using domain knowledge to focus the search on the most promising branches, heuristic search can find good solutions much faster, and algorithms such as A* can still guarantee an optimal solution if the heuristic is admissible (never overestimates the true cost).

Example: Ethiopian logistics and transportation

Consider a logistics company that needs to deliver goods from a warehouse in Addis Ababa to retail shops across the city, or to other cities such as Adama, Hawassa, or Bahir Dar. A heuristic search algorithm such as A* could be used to find the fastest or cheapest delivery routes. The heuristic could be the straight-line (Euclidean) distance to the destination, combined with real-time information such as traffic congestion levels, fuel prices, or road quality. Instead of checking every possible combination of roads (which would be computationally impossible for a large road network), the algorithm uses the heuristic to quickly focus on routes that are likely to be shortest or fastest, helping the company reduce delivery times, fuel costs, and vehicle wear, which is especially valuable given Addis Ababa's traffic congestion and the long distances between major Ethiopian cities.

Question 5: What is overfitting in machine learning? Why is it a problem?

Overfitting occurs when a machine learning model learns the training data too well, including its noise, outliers, and random fluctuations, rather than learning the underlying general pattern. An overfitted model achieves very high accuracy on the data it was trained on, but performs poorly when it encounters new, unseen data, because it has essentially "memorised" the training examples instead of learning rules that generalise.

Overfitting is a serious problem because the ultimate goal of a machine learning model is to make accurate predictions on new, real-world data that it has never seen before, not just on the data used during training. For example, if a model designed to diagnose crop diseases is overfitted to the specific lighting conditions, camera angles, and plant varieties in its training images, it may perform excellently in testing but fail dramatically when used by a farmer with a different phone camera or under different field conditions. This gap between training performance and real-world performance can lead to wasted resources, incorrect decisions, and loss of trust in the system. Overfitting is typically addressed using techniques such as gathering more diverse training data, simplifying the model, using regularisation, and validating the model on data it has not seen during training.

Question 6: What are the ethical issues of using AI in developing countries like Ethiopia? Consider privacy, unemployment, and fairness.

Privacy

Many AI systems rely on collecting large amounts of personal data, such as location data, biometric information (faces, fingerprints), financial records, or health records. In Ethiopia, data protection laws and institutions are still developing, and many people may not fully understand what data is being collected about them or how it will be used. This creates a risk that personal information could be misused, sold, or accessed without proper consent, particularly if foreign companies deploy AI systems with little local regulatory oversight.

Unemployment

AI and automation can replace tasks currently performed by humans, particularly repetitive tasks in agriculture, manufacturing, and customer service, sectors that employ a large share of Ethiopia's workforce. While automation can increase efficiency and economic output, it can also displace workers who may not have access to retraining programs or alternative employment opportunities, potentially widening the gap between skilled and unskilled labour and increasing inequality if the transition is not managed carefully.

Fairness

Many AI models are developed and trained using data from wealthier countries, which may not reflect Ethiopian languages, names, skin tones, agricultural practices, or social contexts. When such models are applied directly in Ethiopia, they may perform poorly or make unfair decisions. For instance, a voice assistant that does not understand Amharic or Afaan Oromo well, or a credit-scoring model trained on data from another country that unfairly disadvantages Ethiopian applicants. Ensuring fairness requires collecting locally representative data, involving local experts in development, and testing systems thoroughly in the Ethiopian context before deployment.

Addressing these issues requires a combination of strong data protection regulations, investment in local AI talent and infrastructure, inclusive policy-making that considers the needs of vulnerable groups, and international cooperation that ensures Ethiopia benefits from AI rather than being left behind or exploited by it.

Question 7: Suppose Addis Ababa City Administration wants to use AI for traffic management. What kind of AI system would you design? Explain your idea.

To address Addis Ababa's growing traffic congestion, I would design an Intelligent Traffic Management System (ITMS) that combines real-time data collection, machine learning-based prediction, and adaptive control of traffic signals and information services.

1. Data collection layer

The system would use a network of cameras and sensors (such as inductive loop sensors or radar) installed at major intersections and along key corridors (for example, around Meskel Square, Bole Road, and the ring road). These would continuously capture data on vehicle counts, speeds, and queue lengths. GPS data from public buses, taxis, and ride-hailing apps could also be incorporated to estimate traffic flow across the wider city.

2. Prediction and analysis layer

Machine learning models (for example, time-series models or neural networks) would be trained on this historical and real-time data to predict congestion levels at different times of day, identify recurring bottlenecks, and detect unusual events such as accidents or road closures that cause sudden traffic build-up.

3. Adaptive control layer

Based on these predictions, the system would dynamically adjust traffic light timings at intersections, for example giving longer green phases to roads with heavier predicted traffic, rather than using fixed timings that do not adapt to actual conditions. During major events or emergencies, the system could automatically create "green corridors" for ambulances or important routes.

4. Public information layer

The system would feed real-time congestion information into a mobile app and digital road signs, helping drivers choose less congested routes, and helping the city's transport authority plan bus schedules and identify locations that need new infrastructure, such as additional lanes or pedestrian bridges. Over time, this would reduce travel times, fuel consumption, and air pollution while improving safety for both drivers and pedestrians.

Question 8: If you were asked to build an AI system for diagnosing crop diseases in rural Ethiopia, what steps would you follow from data collection to deployment?

Building a reliable AI system to help farmers diagnose crop diseases requires a carefully planned process from data collection through to deployment and ongoing support. The following steps outline an appropriate approach:

1. Problem definition and stakeholder engagement

Work with agricultural experts, extension officers, and farmers to identify the most common and economically damaging crop diseases in the target region (for example, diseases affecting teff, maize, or coffee), and define what a useful diagnosis output looks like (for example, disease name, severity, and recommended treatment).

2. Data collection

Collect a large number of images of both healthy and diseased plants/leaves under realistic field conditions, including different lighting, angles, growth stages, and phone camera qualities, since the system will be used in real farms, not controlled laboratories. Where possible, also gather contextual information such as soil type, weather conditions, and region.

3. Data labelling and cleaning

Have agricultural experts label each image with the correct disease (or "healthy"), and remove duplicate, blurry, or mislabeled images. Ensuring high-quality, accurately labelled data is one of the most important steps, as the model's performance depends directly on it.

4. Model development

Select an appropriate model architecture, typically a Convolutional Neural Network (CNN) for image classification, and train it on the labelled dataset, using techniques such as data augmentation (rotating, flipping, adjusting brightness of images) to make the model more robust to the variety of conditions it will face in the field.

5. Evaluation and validation

Test the model on a separate set of images it has not seen during training, ideally collected from different farms or regions than the training data, to check that it generalises well and does not simply overfit to the training conditions. Evaluate not just overall accuracy, but also performance on each individual disease, since some diseases may be rarer than others.

6. Deployment considerations

Package the model into a lightweight mobile application that can run on low-cost smartphones, ideally with offline functionality for areas with limited internet connectivity, and provide results in local languages (such as Amharic, Afaan Oromo, or Tigrinya) with simple visual explanations and practical treatment advice that farmers can act on.

7. Pilot testing and feedback

Deploy the system with a small group of farmers and extension workers first, gather feedback on usability and accuracy, and use this real-world feedback to identify weaknesses, such as diseases the model confuses with one another.

8. Continuous monitoring and improvement

After wider deployment, continue to collect new images (with farmer consent) to expand and update the training dataset over time, since new disease strains may emerge or environmental conditions may change, and periodically retrain the model to maintain and improve its accuracy.